



Linear equations

What to do

- 1 Calculate the gradient of the straight line between the following pairs of points.

a (0, 3) and (3, 6) c (-5, -3) and (1, -1) e (-7, 7) and (-5, 7)	b (1, 5) and (4, 2) d (6, 2) and (-3, 0) f (4, 3) and (4, 1)
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- 2 Write the equation for the straight line that has:

a a gradient of 2 and passes through the point (0, -7) b a gradient of -5 and passes through the point (1, 4) c a gradient of $\frac{1}{5}$ and passes through the point (5, -1).	
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- 3 Consider the following equations.

a $y = -2x + 3$ c $x + 2y = 9$	b $y + 3x = -4$
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 - i Write the equation in the form of $y = mx + c$.
 - ii State the gradient of the line.
 - iii State the y-intercept of the line.
- 4 Find the equation of the line containing the given pair of points.

a (0, 3) and (3, 6) c (-5, -3) and (1, -1) e (-5, 5) and (-5, 7)	b (1, 5) and (4, 2) d (5, -2) and (-2, 1) f (2, -3) and (-4, -3)
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- 5 Descriptions of three possible lines are given listed below.

a A line that does not pass through the first quadrant b A line that passes through exactly two quadrants c A line which passes through only one quadrant	- i Decide whether for each of the above such a line exists. Explain. - ii If a line exists, write a possible equation.
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- 6 Does the graph of the straight line with a slope of -2 and a y-intercept of 5 pass through the point (3, -1)?
- 7 **MC** Which of the following equations contains the two points (-3, -2) and (3, 4)?

A $y = x + 1$ C $y = -x - 7$	B $y = 1 - x$ D $y = -x + 1$
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